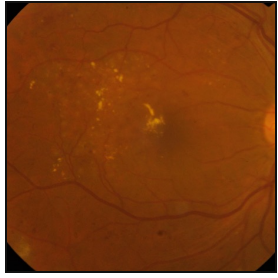
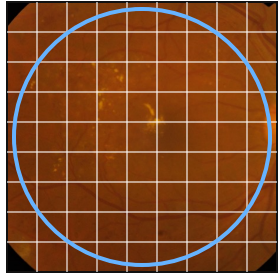


Full-Canvas Canonicalization



Input fundus image
X: $N \times 3 \times H \times W$



Dominant-field crop
+ canonical 896×896

Fixed retinal support mask M_{px} (not a lesion mask)

Spatially Bounded
EfficientNetV2-S
stem—stage 3 • pretrained
stride 8 • RF 95×95
FrozenBN • no SE/GAP

Tap feature map
Ftap: $N \times 64 \times 112 \times 112$

Pointwise Residual MLP
 $64 \rightarrow 128 \rightarrow 256 \rightarrow 128$
no spatial mixing

Local evidence lattice H
 $N \times 12,544 \times 128$
9,864 proof-valid cells

MOSAIC — Minimum Ordinal Sufficient Attribution by Intervention and Counting

Shared Local Ordinal State Head
Linear $128 \rightarrow 5 + \text{softmax}$
 $q(n, i, s) = P(L_i = s)$
 $N \times 12,544 \times 5$

Structurally Nested Witnesses
 $\lambda[n, i, k] = \sum_{(s > k)} q[n, i, s]$
 $\lambda[i, 0] \geq \lambda[i, 1] \geq \lambda[i, 2] \geq \lambda[i, 3]$
 $N \times 12,544 \times 4$

Four Boundary Evidence Maps

dense auxiliary • training only

Exact Truncated Poisson–Binomial Counter
 $Z[i, k] \sim \text{Bernoulli}(\lambda[i, k])$
 $C[n, k] = \sum_{(i \text{ in } V)} Z[i, k]$
 $P(C=0), \dots, P(C=31), P(C \geq 32)$
exact block-tree recurrence • FP32

Dense Evidence
 $cD[n, k] = \sum_r \alpha[k, r]$
 $P(C[n, k] \geq r)$
 $N \times 4$

Learnable Boundary Count Mixtures
 $\alpha[k] = \text{softmax}(\eta[k])$ in $\text{simplex}(32) \cdot \alpha: 4 \times 32$

Deterministic Minimum Top-Prefix Proof
sort $\lambda[i, k]$ descending; select smallest prefix m^*
Sufficiency: $cR[k] \geq cD[k] - \epsilon$
Collective necessity: $cD[k] - cC[k] \geq \rho(cD[k] - \epsilon)$
 $\epsilon = 0.02 \cdot \rho = 0.5$
 $M^*: N \times 12,544 \times 4 \cdot m^*: N \times 4$

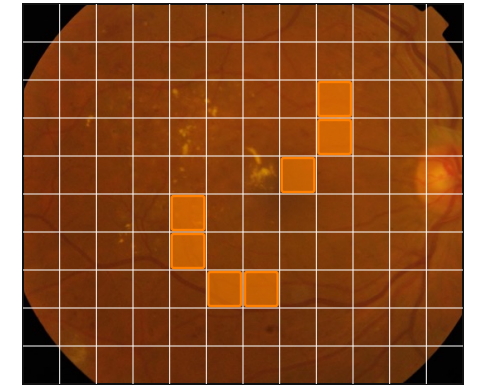
Retained-Proof Circuit Replay
same PB counter on $M^* \times \lambda$
 $c[k] = cR[k]$ and stable $\log s[k]$

Proof-Only Ordinal Cascade
 $P(Y > k) = \text{product}_{(j=0..k)} c[j]$
 $P(Y=0), \dots, P(Y=4)$
 $\hat{y} = \sum_{(k=0..3)} 1[P(Y > k) \geq 0.5]$
raw upper cascade median • parameter-free
5 alternatives: diagnostic audit only

Balanced At-Risk Continuation NLL

$$L_{\text{proj}} = -\sum_{(k < y)} w[k, 1] \log c[k] \\ - 1[y < 4] w[y, 0] \log s[y] \\ L = L_{\text{proj}} + 0.1 L_{\text{dense}} \\ \text{effective-number transition weights}$$

Minimum Ordinal Proof Certificate (inference)



schematic selected witnesses $M^[k]$*

Per boundary: selected indices • proof size $m^*[k]$
dense / retained / complement scores: cD, cR, cC
sufficiency gap • complement drop
optional exact fixed-proof intervention $\Delta[i, k]$
integrity hash + independent numerical replay

Ordinal Outputs (inference)

$$c[0..3] \cdot P(Y > k) \cdot P(Y = 0..4)$$

upper median(q) → predicted grade $\hat{y}$